CODING5S ECOSYSTEM | PEDAGOGY QUICK READ

Sync Student Kit Generator

Automated compilation & localization engine that turns master Creator Kits into production-ready, multilingual Student Kits — without ever touching the source files.

⚙️ PILLAR 1 — CREATOR KITS | 🐍 PYTHON 3.8+ | 🪟 WINDOWS COM AUTOMATION | 🌍 MULTI-LANGUAGE READY

📌 EXECUTIVE SUMMARY

The **Sync Student Kit Generator** is the core distribution engine of the **Coding5s** framework. It recursively scans the ecosystem, opens every master **Creator Kit**, simulates a target language in memory, recalculates all prompts, and freezes the results into clean, protected **Student Kits** — automatically, in the background, at scale.

Built on **Windows COM Automation** (ghost-mode Excel via `DispatchEx`), the pipeline guarantees a **single source of truth**, **zero risk to master files** (`wb.Close(False)`), and **instant global localization** — collapsing hours of manual copy-paste work into a one-click, seconds-long build.

01 WHY THIS GENERATOR MATTERS

🌍 Global Accessibility & Fast Localization

Setting one variable configures every master Creator Kit in memory and compiles a fully localized Student Kit suite — Spanish, Swahili, French, Japanese, German, or any target language — ready for deployment.

🔒 Non-Destructive Creator Kit Protection

The engine updates the language cell, recalculates in RAM, extracts static values, and closes without saving. Master Creator Kits remain 100% untouched and pristine at all times.

🎯 Single Source of Truth

Creators maintain content in one place only. Template cloning, in-memory language updates, cell mapping, and static value freezing are all handled automatically downstream.

⚡ Scalability & Background Execution

Ghost-mode `DispatchEx` runs entirely in the background — no UI flicker, no window interference — turning hours of manual formatting work into a seconds-long automated pass.

02 SYSTEM ARCHITECTURE & COMPONENTS

```
pillar_1/coding5s_creator_kits/
│
├── 📜 Student Kit Template.xlsx      ‹ Master visual template
├── 🐍 student_kit_creator.py         ‹ Core automation engine
├── ⚙️ Student Kit Creator Runner.bat ‹ 1-click execution script
├── 📖 student_kit_creator_guide.md   ‹ Documentation
│
├── 📁 python/  ├── 📁 elixir/  ├── 📁 gleam/  ├── 📁 dart/  ‹ topic subdirectories
```

• Student Kit Template.xlsx

The **master visual template** — pre-styled tables, brand colors, and structural protection.

- Lives beside the automation engine
- Never altered by the script
- Cloned fresh into every subfolder

• student_kit_creator.py

The **core Python engine** — orchestrates discovery, language simulation, extraction, and Excel automation.

- `RESULTS_IN_LANGUAGE` target config
- `GRACIAS` protection password
- `os`, `shutil`, `win32com.client`

• Runner.bat

The **1-click execution script** for Windows users.

- Native VBScript confirmation dialog
- Launches the Python engine directly
- No terminal knowledge required

03 HOW THE BUILD PROCESS WORKS

STAGE 01 Global Template Validation

OBJECTIVE	KEY OUTCOME
Confirm the build can proceed safely before touching any files.	Verifies `Student Kit Template.xlsx` exists in the root folder; aborts safely if missing.

STAGE 02 Recursive Folder Scanning

OBJECTIVE	KEY OUTCOME
Locate every Creator Kit across the entire ecosystem automatically.	`os.walk` scans all subdirectories (python, elixir, gleam, dart…) for filenames containing "Creator Kit," ignoring temp files (~$).

STAGE 03 Isolated COM Initialization

OBJECTIVE	KEY OUTCOME
Run Excel invisibly, with zero interference to the user's desktop.	Launches a background instance via `DispatchEx` — `Visible`, `ScreenUpdating` and `Interactive` all disabled.

STAGE 04 In-Memory Language Simulation & RAM Extraction

OBJECTIVE	KEY OUTCOME
Recalculate every prompt in the target language without saving to disk.	Sets cell B6 on `PromptGenerator`, triggers `excel.Calculate()`, extracts range **E9:Y200** + metadata **I2:I3**, then closes with `wb.Close(False)`.

STAGE 05 Template Cloning

OBJECTIVE	KEY OUTCOME
Produce a clean, unpopulated Student Kit shell for every Creator Kit found.	Target filename replaces "Creator Kit" with "Student Kit"; template is cloned into the subfolder only if it doesn't already exist.

STAGE 06 Desktop Ingestion

OBJECTIVE	KEY OUTCOME
Inject the frozen data into the clean template with a polished final layout.	Unprotects the sheet, injects prompts at **B9** and metadata at **G2:G3**, disables `WrapText`, clears auto-filters.

STAGE 07 Native Save & Structural Lock

OBJECTIVE	KEY OUTCOME
Deliver a secured, distribution-ready file and free system resources.	Re-applies password protection, saves with `wb.Close(True)`, and releases all COM memory objects cleanly.

04 SETUP & REQUIREMENTS

CATEGORY	REQUIREMENT
Operating System	Windows 10 / 11
Software	Microsoft Excel installed locally
Environment	Python 3.8+
Dependencies	`pip install pywin32`

05 STEP-BY-STEP EXECUTION

1. **Set Target Language**
   Open `student_kit_creator.py`, set `RESULTS_IN_LANGUAGE = "English"` (or target), and save.
2. **Verify File Placement**
   Ensure the runner, engine, template, and guide all sit inside `pillar_1/coding5s_creator_kits/`.
3. **Execute the Build**
   Double-click `Student Kit Creator Runner.bat` and confirm the prompt dialog.
4. **Monitor Output**
   The terminal streams real-time status per file: scan path, source/destination pair, and success confirmation.

TECHNICAL FOOTER

FRAMEWORK REFERENCE

🏛️ Key Architectural Pillars

- **Isolated COM sessions** — background Excel via `DispatchEx`
- **RAM-only extraction** — no intermediate files written
- **Template cloning** — shared master, per-module output
- **Recursive discovery** — `os.walk` across all pillars

🛡️ System Guarantees

- **Zero master-file corruption** — always `wb.Close(False)` on Creator Kits
- **Sheet-level protection** restored on every Student Kit output
- **Clean layout** — auto-filters cleared, text-wrap disabled
- **No UI interference** — fully headless, ghost-mode execution

📚 References

- `student_kit_creator.py` — core automation engine
- `Student Kit Template.xlsx` — master visual blueprint
- `student_kit_creator_guide.md` — full documentation
- `Student Kit Creator Runner.bat` — execution entry point